

OSHAWA, ON, Canada

WMO: 716970

Lat: 43.920N

Lon: 78.893W

Elev: 459

StdP: 14.45

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	0.6	-14.3	2.6	-4.0	-8.2	3.6	2.2	26.3	23.4	23.9	18.5	8.3	0	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	85.6	71.0	82.7	69.8	80.4	68.3	74.0	81.5	72.4	79.1	71.0	77.2	10.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	119.2	77.9	70.1	112.7	76.5	68.7	107.4	75.1	37.9	81.8	36.4	79.3	35.2	77.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	20.0	17.9	-12.4	91.2	4.9	4.0	-16.0	94.1	-18.9	96.5	-21.6	98.7	-25.2	101.7		
			-13.0	76.9	5.0	2.1	-16.6	78.4	-19.5	79.6	-22.3	80.8	-25.9	82.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	21.8	22.6	30.7	42.2	56.2	64.6	70.5	68.3	61.6	49.7	37.3	28.8	(6)
(7)		DBStd	19.10	11.36	11.32	10.45	7.46	7.78	5.71	4.93	4.75	7.12	7.76	8.16	9.55	(7)
(8)		HDD50	3677	873	768	603	254	28	1	0	0	2	104	386	658	(8)
(9)		HDD65	7251	1338	1188	1063	685	289	75	10	22	152	475	831	1123	(9)
(10)		CDD50	2337	0	0	6	18	222	439	636	568	348	95	5	0	(10)
(11)		CDD65	436	0	0	0	0	17	63	181	125	49	1	0	0	(11)
(12)		CDH74	3080	0	0	0	2	122	408	1416	797	329	6	0	0	(12)
(13)		CDH80	609	0	0	0	0	12	72	352	116	57	0	0	0	(13)
(14)	Wind	WSAvg	8.2	9.4	9.5	9.6	9.9	7.9	7.5	6.7	6.5	6.7	7.9	8.3	8.5	(14)
(15)	Precipitation	PrecAvg	34.7	2.6	2.1	2.4	3.1	3.0	3.0	3.2	3.0	3.2	3.3	2.8	2.8	(15)
(16)		PrecMax	40.9	5.0	4.3	4.9	5.0	5.6	6.1	6.5	4.9	6.1	6.6	4.4	5.8	(16)
(17)		PrecMin	29.6	1.3	0.9	1.1	1.1	0.9	0.7	1.4	1.1	1.8	1.1	1.2	1.7	(17)
(18)		PrecStd	3.5	1.0	0.8	0.9	1.2	1.3	1.5	1.4	0.9	1.1	1.4	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	51.0	66.1	69.6	81.2	86.1	90.1	87.1	85.4	74.1	61.4	53.8	(19)
(20)			MCWB	46.0	44.7	59.4	56.7	67.0	69.0	73.3	72.5	73.3	63.9	53.4	49.9	(20)
(21)		2%	DB	43.4	45.4	55.0	63.4	77.6	81.8	86.7	82.9	81.2	69.4	56.5	48.0	(21)
(22)			MCWB	41.1	41.5	47.6	52.1	64.7	68.3	72.0	70.7	71.0	62.1	51.1	45.3	(22)
(23)		5%	DB	40.3	41.1	49.2	58.7	73.8	78.5	83.9	80.5	77.8	65.6	53.6	44.4	(23)
(24)			MCWB	38.2	38.2	45.0	50.2	61.7	65.6	69.9	69.0	68.1	59.1	49.0	42.6	(24)
(25)		10%	DB	37.1	37.5	44.9	54.9	70.4	75.5	81.0	78.6	74.5	62.3	50.4	41.3	(25)
(26)			MCWB	35.2	34.7	40.4	47.2	59.6	64.3	68.3	68.1	66.9	57.7	46.4	38.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	45.9	59.2	60.4	69.7	73.4	77.2	75.2	74.8	68.0	55.3	50.9	(27)
(28)			MDCB	48.2	47.3	66.4	67.0	78.8	82.3	87.2	82.0	82.0	69.9	58.8	52.8	(28)
(29)		2%	WB	41.7	42.2	50.4	55.3	66.7	70.7	74.4	73.5	72.5	63.3	52.1	45.8	(29)
(30)			MDCB	42.9	45.2	53.6	58.8	74.5	77.8	83.1	79.2	78.9	67.2	55.4	47.2	(30)
(31)		5%	WB	38.3	38.5	45.5	51.9	64.1	68.7	72.6	71.9	70.6	60.5	49.4	42.8	(31)
(32)			MDCB	40.0	40.9	48.9	57.4	71.8	74.8	80.0	77.5	75.7	64.3	52.7	44.8	(32)
(33)		10%	WB	35.5	34.8	41.0	48.4	61.4	66.7	71.0	70.4	68.2	58.1	46.8	39.1	(33)
(34)			MDCB	37.0	37.3	44.2	53.5	67.5	72.6	77.6	76.0	72.8	61.2	50.2	41.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	15.4	15.3	16.9	18.4	17.7	18.7	17.7	18.0	15.7	14.5	11.7	(35)
(36)			MCDBR	14.1	17.0	20.3	23.7	22.9	21.0	22.0	19.8	20.4	19.7	19.3	13.5	(36)
(37)			MCWBR	13.5	14.6	14.7	15.6	12.3	10.8	10.6	10.2	11.3	13.4	15.6	12.8	(37)
(38)		5% WB	MCDBR	14.8	16.6	18.8	19.8	19.5	17.8	18.7	16.0	18.3	16.7	17.2	13.0	(38)
(39)			MCWBR	14.6	15.0	14.8	15.4	11.7	11.1	9.9	9.4	11.2	12.8	14.8	13.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.311	0.329	0.372	0.402	0.430	0.427	0.415	0.379	0.361	0.317	0.292	(40)	
(41)		taud	2.351	2.335	2.406	2.334	2.299	2.248	2.270	2.292	2.378	2.410	2.490	2.448	(41)	
(42)		Ebn at Noon	259	279	289	283	276	268	267	266	268	257	255	255	(42)	
(43)		Edh at Noon	25	30	32	38	41	43	42	39	33	28	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	427	689	1099	1412	1722	1856	1877	1646	1312	777	479	342	(44)	
(45)		RadStd	42	64	108	148	183	143	138	83	115	72	62	41	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page